Md. Khalequzzaman Sarker Likhon

Dhaka, Bangladesh · khalequzzamanlikhon@gmail.com · +880 1711 906648
Portfolio · GitHub · LinkedIn · ORCID

RESEARCH INTERESTS

Computer Vision · Large Language Models · Multimodal & Vision-Language Models · Medical & Biomedical Imaging · Trustworthy AI

EDUCATION

Ahsanullah University of Science and Technology Apr 2016 – Jan 2021
B.Sc. in Computer Science and Engineering (CGPA: 2.74/4.00) *Dhaka, Bangladesh*

- Thesis: *Design of an Arrhythmia Classification Algorithm Using 2-D Convolutional Neural Networks*; supervisor: Prof. Dr. Mohammad Shafiul Alam.
- Coursework: Artificial Intelligence, Pattern Recognition, Digital Image Processing, Soft Computing, Data Structures and Algorithms.

RESEARCH MANUSCRIPTS

* *denotes equal contribution*

- R. Tanvir*, M. K. S. Likhon*. Joint Prediction of Ultra-High-Performance Concrete Properties Using a Feature-Tokenizer Transformer with Masked Multi-Task Learning. *Under review, Applied AI Letters (Wiley)*, 2026.
- M. K. S. Likhon, M. M. Sarker. Blur Sensitivity and Generator Dependence of Fingerprint Alteration Detectors Trained on SOCOFing. *Manuscript prepared for Image and Vision Computing (Elsevier)*, 2026.
- M. K. S. Likhon, M. M. Sarker. Data Audit and Revised Baselines for Structural Damage-Type Classification on PEER Hub ImageNet. *In preparation*, 2026.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer, Accelx Inc. Jan 2025 – Present
Video understanding, vision-language models, grounded LLM systems *Dhaka, Bangladesh*

- Fine-tuned a Kinetics-pretrained R(2+1)D-18 for violence recognition and deployed it for real-time multi-camera monitoring; on RWF-2000 it beat optical-flow, pose-fusion, and pose-Transformer models.
- Audited pose-based violence models: their "contact" cues came from keypoints missing for both people, and without identity tracking across frames they could not tell who did what.
- Built frozen-OpenCLIP fall and violence recognizers; traced false positives on real footage to viewpoint, lighting, and posture, and turned them into hard-negative prompts (a training-free fix).
- Built a grounded-alert pipeline: a local LLM writes alerts only from code-computed facts and fine-tuned RoBERTa risk scores, with every citation automatically checked.
- Designed an OCR–CLIP–Qwen3-VL cascade that calls the VLM only for hard cases, checks findings against a knowledge base, and flags weak evidence for review.
- Fine-tuned a YOLOv10-L weapon detector for live video streams.

SELECTED RESEARCH AND ENGINEERING PROJECTS

Telemetry vs. video for automated after-action review [GitHub] Independent study, 2026

- Built a reproducible pipeline where a Video LLM writes after-action reviews of human–human Overcooked play from the event log or gameplay video; every timestamped claim is checked against the log, with random-time and player-swap controls.
- On 15 episodes (three Qwen-VL settings, seven input conditions), log-based claim precision was 67–93% vs. 22–43% for video, and video timing was at or below the random-time baseline. A 1,646-item probe over 76 episodes traced the failure to perception: models read the clock and score at 100% but not what a chef holds (24–26% vs.

25% chance, Qwen2.5-VL). Remaining 61 episodes and blind human ratings pending.

DocuVision: key–value extraction from scanned documents [GitHub] 2026

- Fine-tuned LayoutLMv3-large on FUNSD with a learned question-to-answer linker: strict pair F1 rose from 0.23 (heuristic matcher) to 0.74 with ground-truth words; fuzzy pair F1 rose from 0.15–0.17 to 0.43–0.46 end to end with Tesseract or EasyOCR on raw scans. Model released on Hugging Face.
- Ablations: the linker was the bottleneck (a rule-based linker capped at 0.72 pair F1 even with gold tags), and grouping words in reading order cost 25 F1 points on two-column forms.

DocuRoute: retrieval and text-to-SQL routing [GitHub] 2026

- Built an LLM router for SEC 10-K questions that picks hybrid retrieval (BM25 + dense, rank fusion) or sandboxed text-to-SQL and returns cited answers.
- Built an evaluation harness (context precision/recall, faithfulness, answer relevancy); on a 15-question golden set from one company's filings, a general-domain reranker cut faithfulness from 0.93 to 0.77, so hybrid retrieval is the default.

Semantic segmentation on CamVid: FCN, attention U-Net, DeepLabV3+ [GitHub] 2024

- Compared three ResNet-50-based models on CamVid (32 classes): FCN with a custom transposed-convolution decoder, attention U-Net, and DeepLabV3+; trained with a Dice–Jaccard loss (0.7/0.3) and augmentation.
- mIoU: 34.14% (FCN), 45.72% (attention U-Net), 38.13% (DeepLabV3+). Small batch size due to limited GPU memory; hyperparameter search and CRF post-processing left as future work.

Sentinel: tracking-based video anomaly detection [GitHub] 2026

- Combined YOLOv8n detection, BoT-SORT tracking, and rule-based event logic for falls, loitering, abandoned objects, crowding, and wrong-way motion; 19.7 FPS end to end on an RTX A5000.

Other projects: speech-to-speech translation with voice cloning (components chosen by FLEURS evaluation; a 100 MB CPU model beat NLLB-200 3.3B on German and Japanese) [GitHub]; SentinelRAG (retrieval and hallucination auditing); Research-to-Code (multi-agent coding with human approval); Jarvis (local voice assistant); more on the portfolio project list.

TECHNICAL SKILLS

Methods: video action recognition, pose estimation and tracking, object detection, semantic segmentation, vision-language models and Video LLMs, document understanding, transformer fine-tuning, LLM grounding and RAG, evaluation design, ablation and error analysis
Frameworks: PyTorch, Hugging Face Transformers, OpenCLIP, Ultralytics, OpenCV, scikit-learn, Weights & Biases
Engineering: Python, C/C++, SQL, Bash; FastAPI, TorchServe, Docker, Redis, Ollama, Git, Linux

LANGUAGES

Bengali (native), English (full professional proficiency)